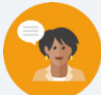


12.2 Finding an Unknown Fraction in an Equation

Lesson Introduction

 Benchmark	MA.6.AR.2.4 Determine the unknown decimal or fraction in an equation involving any of the four operations, relating three numbers, with the unknown in any position.		 Learning Target	I can determine an unknown fraction in a numeric equation with three numbers.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward N/A	
 Essential Questions	- How can you use a bar model to solve an equation? - How can you use reasoning to solve an equation? - Why do different equations have the same solution?			
 Lesson Narrative	Students will solve equations with fractions in an equation involving any one of the four basic operations relating three numbers with the unknown on either side of the equal sign in any position. Students will use algebraic reasoning, drawings, and mental math to solve equations. The terms and solutions will be positive rational numbers.			
 Component Summary	12.2.1 Warm-Up (~5 min.)	How Much Rain? (SE p. 287)	12.2.4 Guided Practice (10 min.)	Solve equations with fractions (SE p. 291)
	12.2.2 Guided Instruction (15 min.)	Use diagrams to solve equations with fractions (SE p. 288)	12.2.5 Your Turn! (5-10 min.)	Practice solving equations with fractions (SE p. 292)
	12.2.3 Collaboration (10 min.)	Solve equations with fractions using reasoning and mental math (SE p. 290)	12.2.6 Wrap-Up (~5 min.)	Represent and solve a word problem using a visual and equation (SE p. 292)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms</u> : N/A <u>Additional Terms</u> : N/A		(Optional) Fraction tiles	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may think they should use the operation and numbers presented in the equation rather than thinking about finding the value of the unknown. - Students may believe coefficients are added to the variable rather than the multiplicative relationship of coefficient and variable. - Students may add or subtract across numerators and denominators rather than finding a common denominator or thinking about the values represented.	- Probing questions to help students connect words to algebra will foreshadow future lessons and get students thinking about their relationships. Do not focus on keywords; instead, focus on the action of the story of the problem and how that action may be represented algebraically. - This lesson is an opportunity to further build students fluency with the operations of rational numbers. Students should not have access to calculators for this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	MLR 6: Three Reads 12.2.2 Guided Instruction uses diagrams to solve equations with fractions. After presenting a problem, ask students to read it several times. Then encourage students to verbally explain what information is given in the problem and what information they are solving for. Also, have students verbally explain how the diagram specifically relates to the problem.	MA.K12.MTR.2.1: Multiple Representations Several activities provide opportunities for students to represent problems in multiple ways. In 12.2.3 Collaboration, encourage students to think about multiple ways to represent each problem. Some students may benefit from drawing a bar model or other visual representation of each problem, while other students may prefer to just work with the numbers. Encourage students to explain why they are using a specific approach to solve a problem.
 Differentiate Instruction	The On-Ramp to 6 th Grade video and practice tool offers a variety of previously learned skills instruction and practice to support all students. Consider having to students preview this lesson by watching the following On-Ramp videos, or have the videos available for students to opt-in throughout this lesson. - Fractions in Real-World Multiplication Problems - Compose and Decompose Fractions - A Fraction Is a Quantity Formed by It's Parts	
Lesson Notes:		

12.2.1 Warm-Up: How Much Rain?

Component Overview: Students use the rain gauges to determine the difference in inches or centimeters in the rain amounts before and after a rain shower. Students will also compare their answer with a partner. Encourage students to explain how they determined their answer.

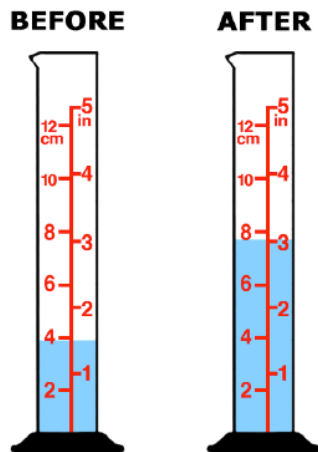


12.2.1 Warm-Up: How Much Rain?

José checked his rain gauge before and after a rain shower.

- How much rain fell during the rain shower?

1.5 inches or 3.9 cm



- Use the pictures to describe how you determined the amount of rain that fell during the rain shower.

Answers will vary. Find the difference between the rain amounts in inches and/or cm on the two gauges.



Review the word “gauge” with students as needed. Although this word is not imperative to solving the problem, students that do not know its meaning may spend too much time on what they do not know.

12.2.2 Guided Instruction: Using Diagrams to Solve Equations With Fractions

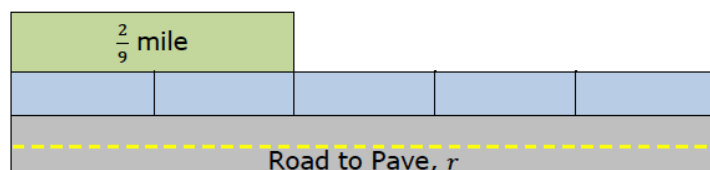
Component Overview: Students use models to find the unknown value in equations with fractions. The activity introduces a variety of models such as bar diagrams, drawings, and measurement devices. In addition to solving for an unknown, students will also explain how they arrived at their answers.



12.2.2 Guided Instruction: Using Diagrams to Solve Equations with Fractions

Diagrams are helpful tools when solving equations for unknown values.

1. A road construction crew paved $\frac{2}{9}$ mile, which is $\frac{2}{5}$ of a full paving job. The bar model represents the equation $\frac{2}{9} = \frac{2}{5}r$, where r is the length of the whole road being paved.



- a. Discuss with your partner how the diagram represents the information given.

Answers will vary. The bottom and middle bars represent the entire amount to be paved. The top bar shows the amount that has already been paved.

- b. Describe how to use the bar model to solve the equation for r , the length of the road being paved.

Answers will vary. Use the top bar to figure out the length of the unpaved section and then add the amounts together.

- c. How long is the road being paved?
 $\frac{5}{9}$ mile

- d. Check your solution. $\frac{2}{9} = \frac{2}{5} \left(\frac{5}{9} \right)$



Encourage students to verbally explain the problem. Have students tell a partner what information is given and what information they need to find. Discuss how to use the bar model and if another approach may also work to solve the problem.



Scaffold instruction through intentional questioning. $\frac{2}{9}$ mile is $\frac{2}{5}$ of the entire paving job.

- Why do we use multiplication, $\frac{2}{5}r$, in the equation?
- How can we use the drawing to help us find the solution?
- Does your answer make sense?
- How can you check your solution?

12.2.2 Guided Instruction: Using Diagrams to Solve Equations With Fractions (continued)

2. Shirlene had $4\frac{1}{8}$ yards of ribbon. She decided to use some for her craft project. She has $2\frac{3}{4}$ yards of ribbon left.

The model represents the amount of ribbon Shirlene had before she started her craft project.



The equation $4\frac{1}{8} - x = 2\frac{3}{4}$ can be used to find x , the amount of ribbon used for the craft project.

- Shade the part of the model that represents the amount of ribbon Shirlene has left after her craft project.
- What does the unshaded part of the model now represent?

The unshaded part of the model represents the amount of ribbon Shirlene used for her craft project.

- Use the model to find the value of x in the equation $4\frac{1}{8} - x = 2\frac{3}{4}$.

$$x = 1\frac{3}{8} \text{ yards}$$

- How much ribbon did she use for the craft project?
 $1\frac{3}{8} \text{ yards}$

- Check your solution. $4\frac{1}{8} - 1\frac{3}{8} = 2\frac{3}{4}$



Discuss why the equation involves subtraction. Have students verbally explain how to use the bar model to solve the problem. Have students show their partner $2\frac{3}{4}$ on the bar model. Ask students if they can use the bar model another way.



Rephrase equations with questions, substituting “what number” in the place of the variable. For example, “ $4\frac{1}{8}$ minus what number will equal $2\frac{3}{4}$,” or “What number could you subtract from $4\frac{1}{8}$ to get $2\frac{3}{4}$?” This will deepen student understanding of what a variable is and represents in an equation.



To challenge students, pose the following questions:

- How could you solve the problem without using the bar model?*
- Could you create a different model to represent the problem?*

12.2.2 Guided Instruction: Using Diagrams to Solve Equations With Fractions (continued)

3. Donté used milk for 3 batches of a recipe. He used $\frac{3}{4}$ cup of milk for each batch.



MA.K12.MTR.2.1: Multiple Representations

Encourage students to think about different ways they could represent this problem. Some students may benefit from drawing a bar diagram. Also ask students if they can think of multiple ways to mathematically solve the problem. Have students discuss the different possibilities of each approach along with the strengths and weaknesses of each.

The equation $m \div 3 = \frac{3}{4}$ can be used to find m , the amount of milk Donté used.

- a. Use the measuring cups shown to solve the equation $m \div 3 = \frac{3}{4}$.

$$\frac{m}{3} = \frac{3}{4}$$

$$m = (3) \frac{3}{4} = \frac{9}{4} = 2\frac{1}{4}$$

- b. Check your solution. $2\frac{1}{4} \div 3 = \frac{3}{4}$

12.2.3 Collaboration: Solving Equations With Fractions Using Reasoning and Mental Math

Component Overview: Students will work together to solve one-variable equations with fractions. Encourage students to use reasoning and mental math to solve the problems, but allow students to draw diagrams as needed. Students will also note equations that have the same solution and discuss how that could occur.



12.2.3 Collaboration: Solving Equations with Fractions Using Reasoning and Mental Math

1. Solve each equation. Check each of your solutions.

$$\frac{2}{3} \cdot d = \frac{10}{9}$$

$$d = 1\frac{2}{3}$$

$$e \div \frac{5}{3} = 1$$

$$\frac{e}{\frac{5}{3}} = 1$$

$$e = 1 \cdot \left(\frac{5}{3}\right)$$

$$e = 1\frac{2}{3}$$

$$\frac{1}{2} = 3\frac{1}{4} - w$$

$$w = 2\frac{3}{4}$$

$$1 = h + \frac{3}{7}$$

$$-h = \frac{3}{7} - 1$$

$$-h = \frac{3-7}{7}$$

$$-h = \frac{-4}{7}$$

$$h = \frac{4}{7}$$

$$\frac{1}{4}f = \frac{1}{7}$$

$$f = \frac{1}{7} \cdot \frac{4}{1}$$

$$f = \frac{4}{7}$$

$$x - 1\frac{1}{2} = 1\frac{1}{4}$$

$$x = 2\frac{3}{4}$$



Encourage students to draw diagrams to represent each problem as needed. Some students may also benefit from creating a story to explain each problem. For example, “Juan has $\frac{3}{7}$ of a loaf of bread. How much more bread does he need to make a whole loaf?” Making a problem more concrete often helps students who feel overwhelmed or bewildered by an abstract problem with a variable.



Consider having students share the strategies they use to do the mental math. Help students use mathematical language in their description. Avoid phrases such as “keep, change, flip”. Encourage phrases such as “multiply by the reciprocal”. Accept any equivalent response as correct. Help students perform the mental math to determine if their answer is equivalent.

12.2.3 Collaboration: Solving Equations With Fractions Using Reasoning and Mental Math (continued)

2. Some of the equations have equivalent solutions. Ask two classmates for a pair of equations that have equivalent solutions. In the table, record their solution and write the equations they tell you gave them that solution. *You are the only person who should write in the first two columns of the table.* Have the person initial next to the equations they told you have equivalent solutions.

Solution	Equations	Initials
	<i>Answers will vary. Any equations that have the same solution are correct.</i>	



Challenge students to create a pair of equations that have the same solution. For more of a challenge, ask students to use different operations in the equations.

12.2.4 Guided Practice: Solving Equations With Fractions

Component Overview: Students will work to solve one-variable equations with fractions. Some students may need to be guided through the activity step by step, while other students will feel confident about working through the questions independently.



12.2.4 Guided Practice: Solving Equations with Fractions

1. Raul made jam and filled four equal-sized jars with $2\frac{2}{3}$ cups of jam. The equation $j \div 4 = 2\frac{2}{3}$ can be used to find the amount of jam, j , Raul made.

- a. Draw a diagram that represents this situation.



- b. How much jam did Raul make? Include units.
 $10\frac{2}{3}$ cups

2. The equation $2 = \frac{11}{9} + n$ can be solved using reasoning.

- a. How many ninths are equal to 2?
18

- b. Explain how the answer to the previous question can be used to reason about the value of n .

You can use the difference between $\frac{18}{9}$ and $\frac{11}{9}$.

- c. What is the value of n ?
 $n = \frac{7}{9}$



If students struggle with a bar diagram, encourage them to draw pictures instead of a bar diagram. Encourage thinking with number sense and reasoning about what value could be placed in the place of the variable to make the equation true.

Monitor students as they work through the guided practice. If students are struggling with determining the answer, then consider working with these students in small groups and have them verbalize their thinking. Consider modeling by verbalizing your thinking.

12.2.5 Your Turn!

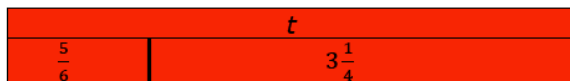
Component Overview: Students will practice solving one-variable equations with fractions. Students may engage with this activity independently or peer review a partner's responses.



12.2.5 Your Turn!

1. Delia had a piece of decorative border. She cut $\frac{5}{6}$ feet off to make the piece $3\frac{1}{4}$ feet. The situation is modeled by the equation $t - \frac{5}{6} = 3\frac{1}{4}$, where t is the length in feet of the piece of trim before it was cut.

- a. Draw a diagram that represents this situation.



- b. How long was the piece of border before Delia cut it? Include units.

$4\frac{1}{12}$ feet



If you notice students struggling with this section as they work independently, pose questions like:

- Why does the equation involve subtraction?
- How can you create a fraction model to find the solution?
- How can I show $3\frac{1}{4}$ on the fraction model?
- How is $3\frac{1}{4}$ related to $\frac{5}{6}$?
- Does your answer make sense? How can you check your solution?

12.2.6 Wrap-Up: Ticket Out the Door

Component Overview: Students create a bar diagram to model the situation described in the problem. Students then use the diagram to determine how much water was dispensed from the bottle. Students should not use a calculator to complete the Wrap-Up.



12.2.6 Wrap-Up: Ticket Out the Door

1. Mr. Bing placed a new 5-gallon water bottle in the office water dispenser when he arrived in the morning. At the end of the day, the bottle had $2\frac{7}{8}$ gallons remaining. The equation $5 = x + 2\frac{7}{8}$ can be used to find x , the gallons of water dispensed from the bottle.

- a. Draw a diagram representing the situation.



- b. How much water was dispensed from the bottle?
Include units.

$2\frac{1}{8}$ gallons



Students may need a quick review or explanation of what the word “dispensed” means to clear up any contextual stumbling blocks to finding the answer.

Use the information from the ticket out the door and class observations to inform further instruction and small group instruction. Consider whether the student is struggling determining the correct operation or if the student is struggling with applying mental math to operations with fractions. If students show work for the addition of the fractions, then consider adding mental math moments to build student fluency.